

Solution — GIAM 5.2 Problem 5

Problem

Prove the following formula for a product:

$$\prod_{i=2}^n \left(1 - \frac{1}{i}\right) = \frac{1}{n}, \quad n \geq 2.$$

Direct Proof (Telescoping)

Rewrite each factor as a single fraction:

$$1 - \frac{1}{i} = \frac{i-1}{i}.$$

Therefore

$$\prod_{i=2}^n \left(1 - \frac{1}{i}\right) = \prod_{i=2}^n \frac{i-1}{i} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{n-1}{n}.$$

Every numerator after the first cancels the denominator of the previous factor (a telescoping product), leaving

$$\prod_{i=2}^n \left(1 - \frac{1}{i}\right) = \frac{1}{n},$$

as claimed.

Proof by Induction

Basis ($n = 2$):

$$\prod_{i=2}^2 \left(1 - \frac{1}{i}\right) = 1 - \frac{1}{2} = \frac{1}{2} = \frac{1}{n}.$$

Inductive step. Assume, for some integer $k \geq 2$,

$$\prod_{i=2}^k \left(1 - \frac{1}{i}\right) = \frac{1}{k}.$$

Then

$$\begin{aligned} \prod_{i=2}^{k+1} \left(1 - \frac{1}{i}\right) &= \left[\prod_{i=2}^k \left(1 - \frac{1}{i}\right) \right] \cdot \left(1 - \frac{1}{k+1}\right) \\ &= \frac{1}{k} \cdot \frac{k}{k+1} \\ &= \frac{1}{k+1}. \end{aligned}$$

By the principle of mathematical induction, the identity holds for all integers $n \geq 2$.